DistillSemiCD: DINOv2-Driven Distillation for Semisupervised Remote Sensing Change Detection

Yetong Xu, Yueping Huang, Jingjing Li, Yang Zhao, Tao Lei, *Senior Member, IEEE*, and Ganchao Liu

***Abstract*—Although semisupervised learning (SSL) can effectively reduce the model's dependence on large amounts of precisely annotated data for remote sensing change detection (RSCD), existing methods still face several challenges. First, existing lightweight architectures struggle to extract discriminative semantic features under limited supervision, while vision foundation models (VFMs) are computationally prohibitive for practical deployment. Second, the inherent noise in pseudolabels tends to accumulate and propagate during training, thereby degrading model performance. Third, changing boundaries frequently exhibit low confidence and blurred localization. To address these challenges, we propose DistillSemiCD. First, we construct a train–inference decoupled knowledge distillation (TID-KD) framework, where a frozen DINOv2 serves as the teacher during training to transfer high-level semantic knowledge to a lightweight student model. During inference, only the student model is deployed, achieving efficient performance comparable to DINOv2. Second, we propose a DINO-based adaptive pseudolabel calibration (DINO-APC) module and a dual-level contrastive learning (DLCL) mechanism. DINO-APC leverages VFM priors to dynamically adjust pseudolabel confidence for noise filtration, while DLCL suppresses noise propagation and feature distribution shift through pixel-level local contrast and batch-level class center alignment. Finally, we design a Taylor-inspired boundary refinement (TIBR) module that employs dual attention gates to localize uncertain regions and uses a discrepancy-guided feature refinement mechanism to sharpen prediction boundaries. Extensive experiments on the WHU-CD, SYSU-CD, CDD-CD, and LEVIR-CD datasets demonstrate that DistillSemiCD achieves competitive performance across diverse benchmark scenarios. The code will be publicly available at https://github.com/Sean1005-x/DistillSemiCD**

***Index Terms*—Knowledge distillation (KD), remote sensing change detection (RSCD), semisupervised learning (SSL), vision foundation models (VFMs).**

I. Introduction

REMOTE sensing change detection (RSCD) aims to identify the dynamic evolution of land surface coverage by quantifying spatiotemporal differences between bitemporal images. This technology serves as a critical pillar in domains such as urban planning monitoring [1], disaster emergency management [2], precision agriculture [3], and national defense security [4]. Although fully supervised learning (FSL) methods based on deep learning have achieved dominance in change detection due to their powerful feature extraction capabilities [5], [6], [7], [8], their performance heavily relies on massive datasets with high-quality annotations. Fine-grained annotation of bitemporal remote sensing imagery requires domain experts to perform pixelwise interpretation, which is prohibitively time-consuming and laborious, severely constraining the practical applications of FSL methods. To circumvent the annotation burden, unsupervised learning (UL) methods have been explored for change detection [9]. However, these approaches suffer from limited detection accuracy due to the lack of explicit supervision signals. Semisupervised learning (SSL) offers an ideal solution to alleviate the annotation bottleneck by effectively leveraging massive unlabeled data with only limited labeled samples [10], and has been thoroughly validated in domains such as medical image segmentation [11] and autonomous driving scene understanding [12].

Existing semisupervised methods primarily exploit unlabeled data through four strategies: adversarial training, consistency regularization, pseudolabeling, and contrastive learning. Adversarial methods utilize the adversarial learning mechanism between generators and discriminators to align data distributions [13]. However, these methods suffer from training instability, mode collapse, and hyperparameter sensitivity, hindering their robustness in complex remote sensing scenarios. Consistency regularization methods follow the smoothness assumption, enforcing prediction consistency under perturbations [14]. Consistency constraints can be implemented through data augmentation [15], [16], feature perturbation [17], [18], or model perturbation [19], [20]. However, these methods exhibit high sensitivity to perturbation intensity. When bitemporal images contain significant illumination differences or seasonal variations, excessive perturbation destroys feature semantics, while insufficient perturbation prevents the model from effectively learning robust semantic representations. Pseudolabeling approaches expand the training set by

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iteratively generating high-confidence pseudolabels [21], [22], but suffer from severe confirmation bias. In change detection tasks, class imbalance and complex land cover appearances cause models to produce erroneous pseudolabels in ambiguous boundary regions or high-uncertainty areas. These noisy labels are continuously reinforced during self-training, leading to error accumulation and performance degradation. Contrastive learning methods pull positive samples closer while pushing negatives apart in the feature space [23], [24], enhancing feature discriminability. However, they heavily rely on high-quality sample pairs. In change detection, where changed regions are typically sparse and irregular, designing effective sampling strategies to handle class imbalance remains challenging.

Recent studies have combined the aforementioned strategies, such as the integration of pseudolabeling with consistency regularization [25], [26], to enhance model performance. Despite certain progress, three critical challenges persist in complex RSCD tasks.

1) *Contradiction Between Representation Capacity and Deployment Efficiency:* Most existing methods rely on lightweight networks (e.g., ResNet-50), which struggle to learn discriminative features when annotations are scarce, leading to high false alarm rates in complex scenes. Although vision foundation models (VFMs) such as DINOv2 [27] offer robust generalizable representations that could mitigate this issue, their substantial parameter counts and computational overhead preclude deployment in resource-constrained scenarios. Recent studies have begun exploring this direction: ChangeViT [28] demonstrates that DINOv2-pretrained ViTs exhibit stronger large-scale change perception than CNN counterparts, and UniMatch V2 [29] verifies that replacing ResNet with DINOv2 yields substantial gains in semisupervised dense prediction. However, dedicated semisupervised RSCD frameworks that jointly optimize DINOv2-based knowledge transfer, pseudolabel calibration, and training–inference decoupling remain scarce.
2) *Pseudolabel Noise Accumulation and Confirmation Bias:* In class-imbalanced scenarios, models tend to overfit easy samples while generating erroneous pseudolabels for hard samples (e.g., boundary-ambiguous regions). This noise propagates through iterative training, exacerbating confirmation bias. Existing consistency constraints are primarily applied at the prediction level, failing to correct such bias at the feature distribution level, which results in intraclass dispersion and interclass confusion in the feature space.
3) *Insufficient Boundary Refinement Capability:* Low-confidence regions in semisupervised settings are typically concentrated at change boundaries. Existing approaches often lack mechanisms to handle boundary uncertainty, resulting in suboptimal segmentation, particularly when labeled data is limited.

To address these challenges, we propose DistillSemiCD, a training–inference decoupled semisupervised change detection distillation framework. We posit that the robust semantic representations of large-scale VFMs are primarily necessary during the training phase to guide pseudolabel generation and feature learning, whereas the inference phase can be independently executed by a fully distilled, lightweight student model. In particular, we employ a frozen DINOv2 as the teacher model and transfer robust semantic knowledge to a lightweight student network (e.g., ResNet) through spatial feature distillation. To suppress noisy labels, we leverage DINOv2's sensitivity to structural information for calibrating pseudolabel confidence and introduce dual-level contrastive learning (DLCL) to increase interclass distances in the feature space. For boundary ambiguity issues, we design a boundary iterative refinement module inspired by Taylor's finite difference to explicitly recover boundary details.

Our main contributions are summarized as follows.

1) To address the conflict between model efficiency and performance, we propose the train–inference decoupled knowledge distillation (TID-KD) framework. During training, a frozen DINOv2 serves as the teacher model to guide a lightweight student network via spatial feature alignment, facilitating the learning of robust semantic representations. During inference, only the student network is retained, thereby achieving an optimal trade-off between computational efficiency and segmentation performance.
2) To mitigate pseudolabel noise accumulation, we propose DINO-APC and DLCL. DINO-APC exploits the prior knowledge of DINO to adaptively calibrate confidence scores for noise filtration, while DLCL prevents feature distribution shift through dual alignment constraints at both pixel-level and batch-level class prototypes. These two components constitute a local–global collaborative antinoise mechanism, significantly enhancing the model's robustness against noisy labels.
3) To address low accuracy in boundary regions, we design the TIBR. This module localizes high-uncertainty boundary regions through a dual-attention gating mechanism and introduces a learnable discrepancy-guided iterative update rule in feature space to progressively sharpen and refine boundary features, effectively alleviating boundary information loss and spatial localization deviation.

II. Related Works

A. Semisupervised Learning

Mainstream SSL methods can be grouped into four categories: adversarial training, consistency regularization, pseudolabeling, and contrastive learning.

Early adversarial training methods, such as s4GAN [30], pioneered the integration of a discriminator following the segmentation network, utilizing the discriminator's output as a confidence map to select high-quality unlabeled samples for training. Building upon this adversarial paradigm, ADS-SemiSeg [31] further proposed an adversarial dual-student framework, which introduces two independently trained student models with stability constraints, leveraging the discrepancies between models to enhance generalization capabilities.

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The paradigm of consistency regularization enforces prediction invariance under input or model perturbations. FixMatch [32] applied weak and strong augmentations to the same sample and forced the prediction on the strongly augmented view to align with the pseudolabel generated from the weakly augmented view. Building on this, UniMatch [33] introduced a dual-stream perturbation strategy at both image and feature levels to significantly improve feature robustness. Meanwhile, iMAS [34] addressed interinstance heterogeneity by proposing instance-specific and model-adaptive supervision strategies, dynamically adjusting augmentation intensity based on sample confidence.

Pseudolabeling methods involve iterative self-training by generating pseudolabels. ST++ [35] refined the classic self-training pipeline by employing multistage selective retraining on unlabeled images subject to strong data augmentation. CPS [36] adopted a cross-supervision strategy where two networks with identical structures but different initializations supervise each other, utilizing network divergence to expand the exploration of the solution space. U2PL [37] focused on the utilization of uncertainty in pseudolabels, storing low-confidence pixels in a negative sample queue rather than discarding them, thereby maximizing data utilization efficiency.

Contrastive learning-based methods aim to enhance feature discriminability. For instance, SemiSeg [38] combined label space consistency with pixel-level contrastive loss, addressing the false negative problem through an improved negative sampling mechanism to bolster intraclass compactness. ReCo [39] extended contrastive learning to the region level, applying pixel-level contrast regarding sparse hard negative samples to guide the model in learning sharper interclass boundaries.

However, existing methods struggle with pseudolabel noise accumulation in semisupervised settings. When directly transferred to RSCD, their performance is severely limited by significant domain gaps and the absence of tailored feature constraints for small objects and ambiguous boundaries, leading to suboptimal detection in complex scenes.

B. Semisupervised Change Detection

With the rapid advancement of remote sensing imaging technology, SSL has emerged as a promising paradigm to reduce annotation dependency in RSCD. Recent surveys [40], [41] have highlighted SSL as one of the most practical directions, owing to its ability to leverage large amounts of unlabeled bitemporal data while significantly reducing annotation costs. Early adversarial methods, such as SemiCDNet [42], introduced dual-discriminator architectures with entropy minimization to produce high-confidence change maps from unlabeled data. Consistency regularization methods subsequently became mainstream: SemiCD [43] enforced feature-level consistency on perturbed bitemporal pairs, CutMix-CD [44] diversified change scenarios through change-aware mixed augmentation, LGTC [45] extended consistency to pixel, region, and image levels to strengthen structural continuity and global semantic coherence, and WACS-SemiCD [46] further demonstrated that weak–strong augmentation combined with adaptive CutMix and class-balanced sampling substantially improves model robustness and cross-domain generalization. For pseudolabel refinement, ECPS [47] aggregated multinetwork predictions to balance precision and recall, AdaSemiCD [48] dynamically adjusted pseudolabel weights to suppress early stage noise, and BRT [49] specifically targeted boundary imprecision by introducing a boundary refinement module with a stabilized teacher update strategy to improve the geometric fidelity of unlabeled supervision. Regarding contrastive learning, RCL [50] constructed reliable positive and negative pairs from unlabeled features, FPA [51] combined class-aware feature and prediction alignment to reduce intraclass discrepancy, and SC2A-SCD [52] advanced this direction by jointly modeling classification and representation spaces through confusion-aware contrastive learning to suppress interclass confusion among semantically similar categories. Most recently, visual-language priors have been introduced into SSCD: SemiCD-VL [53] leveraged VLM-generated auxiliary supervision with dual projection heads and contrastive consistency regularization, while SemiBCD [54] further showed that VLM-generated pseudolabels require uncertainty-aware refinement and dual consistency constraints to be reliably exploited. Despite these advances, existing methods still face three critical challenges: 1) limited representational capacity under scarce labels; 2) pseudolabel noise accumulation and propagation; and 3) insufficient boundary localization. These limitations collectively motivate the design of our proposed DistillSemiCD.

C. Knowledge Distillation

Knowledge distillation (KD) aims to transfer knowledge from a teacher network to a student network. Response-based distillation [55] aligns prediction distributions at the output layer, but often loses spatial structural information that is critical for change detection. Feature-based methods, such as AT [56] and IFVD [57], encourage the student to mimic intermediate feature representations; however, they usually treat all regions equally and overlook the semantic imbalance between changed and unchanged areas. Relation-based methods, including SKD [58] and CWD [59], further introduce structural constraints by modeling pixelwise, pairwise, global, or channelwise relationships. With the development of large-scale self-supervised pretraining, VFMs such as DINOv2 [27] have shown strong generalizable representations, making them promising frozen teachers for compact task-specific networks. Nevertheless, this paradigm remains underexplored in RSCD, particularly regarding the joint design of distillation and pseudolabel generation, which our proposed DistillSemiCD addresses through a decoupled training–inference strategy.

D. Boundary Refinement for Change Detection

Boundary refinement has emerged as a critical research direction to address blurred contours, incomplete boundaries, and interior holes in change detection. Under fully supervised settings, B2CNet [60] proposes a progressive boundary-to-center refinement paradigm that enhances both contour precision and region integrity without requiring additional

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edge annotations. AERNet [61] introduces an attention-guided edge refinement module to strengthen building boundary extraction, while EGPNet [62] introduces an edge-guided refinement module to improve target shape integrity while suppressing clutter background interference in infrared imagery. In the semisupervised setting, BRT [49] incorporates a bitemporal boundary refinement module to optimize pseudolabel boundaries and stabilizes the training process via an improved teacher update strategy. Under weakly supervised settings, BARNet [63] designs multiscale boundary attention and boundary-aware decoupling modules to derive precise boundaries from image-level labels alone. These methods consistently demonstrate that explicit boundary modeling significantly improves change detection accuracy across diverse supervision paradigms. Despite these advances, dedicated boundary refinement mechanisms in semisupervised change detection remain limited.

III. Methodology

A. Overview and Pipeline

1) Problem Definition: In the SSL scenario, the labeled data are defined as $\mathcal{D}_L = \{(\mathbf{I}_A^i, \mathbf{I}_B^i, \mathbf{Y}^i)\}_{i=1}^{N_L}$, and the unlabeled data as $\mathcal{D}_U = \{(\mathbf{I}_A^j, \mathbf{I}_B^j)\}_{j=1}^{N_U}$, where $N_U \gg N_L$. Here, $\mathbf{I}_A$ and $\mathbf{I}_B$ represent bitemporal remote sensing images, and $\mathbf{Y} \in \{0,1\}^{H\times W}$ denotes the binary change map.

2) Framework Overview: As illustrated in Fig. 1, the proposed framework operates in two phases with distinct computational paradigms.

1) *Training Phase:* The architecture comprises four key components: 1) a ResNet50-based student encoder E_s that extracts multiscale difference features from bitemporal images; 2) a frozen DINOv2 foundation model E_t serving as the teacher network, providing semantic supervision through spatial feature distillation and pseudolabel calibration; 3) a Taylor-inspired boundary refinement (TIBR) module B that explicitly addresses boundary ambiguity; and 4) a lightweight decoder D employing depthwise separable convolutions for efficient segmentation map generation.
2) *Inference Phase:* Critically, the trained student network operates completely independently of the DINOv2 teacher, requiring only the encoder–decoder architecture ($E_s \to B \to D$) with approximately 54 M parameters. This training–inference decoupling enables foundation model-level performance at the inference cost approximating that of a ResNet, addressing the fundamental deployment challenge of large-scale foundation models.

3) Backbone Architecture: Our student encoder employs ResNet50 as the backbone to extract hierarchical difference features at four scales

$$\mathbf{F}_i = |\phi_i(\mathbf{I}_A) - \phi_i(\mathbf{I}_B)|, \quad i \in \{1,2,3,4\} \tag{1}$$

where $\phi_i(\cdot)$ denotes the feature extraction function of the ith stage of ResNet50, mapping the input image to a spatial feature map.

The deepest features $\mathbf{F}_4$ are processed through an atrous spatial pyramid pooling (ASPP) module to capture multiscale contextual information

$$\mathbf{F}_{\text{ASPP}} = \text{ASPP}(\mathbf{F}_4) = \text{Concat}\left[\mathcal{C}_1, \mathcal{C}_6, \mathcal{C}_{12}, \mathcal{C}_{\text{global}}\right] \tag{2}$$

where $\mathcal{C}_r$ denotes atrous convolution with dilation rate r, and $\mathcal{C}_{\text{global}}$ represents upsampled global average pooling. The concatenated features are projected to 512 channels through a 1×1 convolution.

B. Train–Inference Decoupled KD

1) Frozen DINOv2 Teacher Model: DINOv2 is employed as a frozen teacher model exclusively during the training phase. In the inference phase, the teacher is removed, incurring zero additional computational overhead while enabling the student network to inherit the foundation model's semantic representation capability.

In particular, the teacher model processes the resized bitemporal input images separately to extract semantically rich spatial features

$$\mathbf{F}_t^A = E_t(\text{Resize}(\mathbf{I}_A, 224)), \quad \mathbf{F}_t^B = E_t(\text{Resize}(\mathbf{I}_B, 224)). \tag{3}$$

Features are extracted from the 11th Transformer layer of DINOv2, prior to the final layer normalization. This choice is motivated by the layerwise representation behavior of ViT-based foundation models. Shallow layers mainly preserve local texture and edge cues, whereas very deep layers tend to encode highly global semantic responses. For dense change detection, the teacher feature should simultaneously provide semantic discriminability and preserve token-level spatial structure for pixel-wise alignment. The 11th layer, therefore, offers a suitable tradeoff between high-level semantic abstraction and fine-grained spatial localization. The teacher's different features are computed as

$$\mathbf{F}_t^{\text{diff}} = |\mathbf{F}_t^A - \mathbf{F}_t^B| \in \mathbb{R}^{768\times 16\times 16}. \tag{4}$$

2) Spatial Feature Distillation: To enable the student encoder to inherit the powerful semantic comprehension capability of the teacher, we design a learnable spatial feature translator $\mathcal{T}$ that aligns the student features to the teacher's feature space

$$\mathcal{T}(\mathbf{F}_s) = \text{BN}\left(\mathbf{W}_2 \cdot \text{ReLU}\left(\text{BN}\left(\mathbf{W}_1 \cdot \mathbf{F}_s^{\text{diff}}\right)\right)\right) \tag{5}$$

where $W_1 \in \mathbb{R}^{256\times 512}$ and $W_2 \in \mathbb{R}^{768\times 256}$ are learnable projection weights, and BN denotes batch normalization. Analogously, the student difference feature is defined as $F_s^{\text{diff}} = |F_4^A - F_4^B|$, where F_4^A and F_4^B are the deepest-stage feature maps extracted by the student encoder E_s for the prechange and postchange images, respectively. The spatial distillation loss enforces feature alignment at corresponding spatial locations

$$\mathcal{L}_{\text{distill}} = \text{MSE}\left(\mathcal{T}(\mathbf{F}_s), \mathbf{F}_t^{\text{diff}}\right). \tag{6}$$

This loss encourages the student to learn spatially aware semantic representations, enabling it to discriminate between changed and unchanged regions with foundation model-level discriminative power while maintaining ResNet's computational efficiency.

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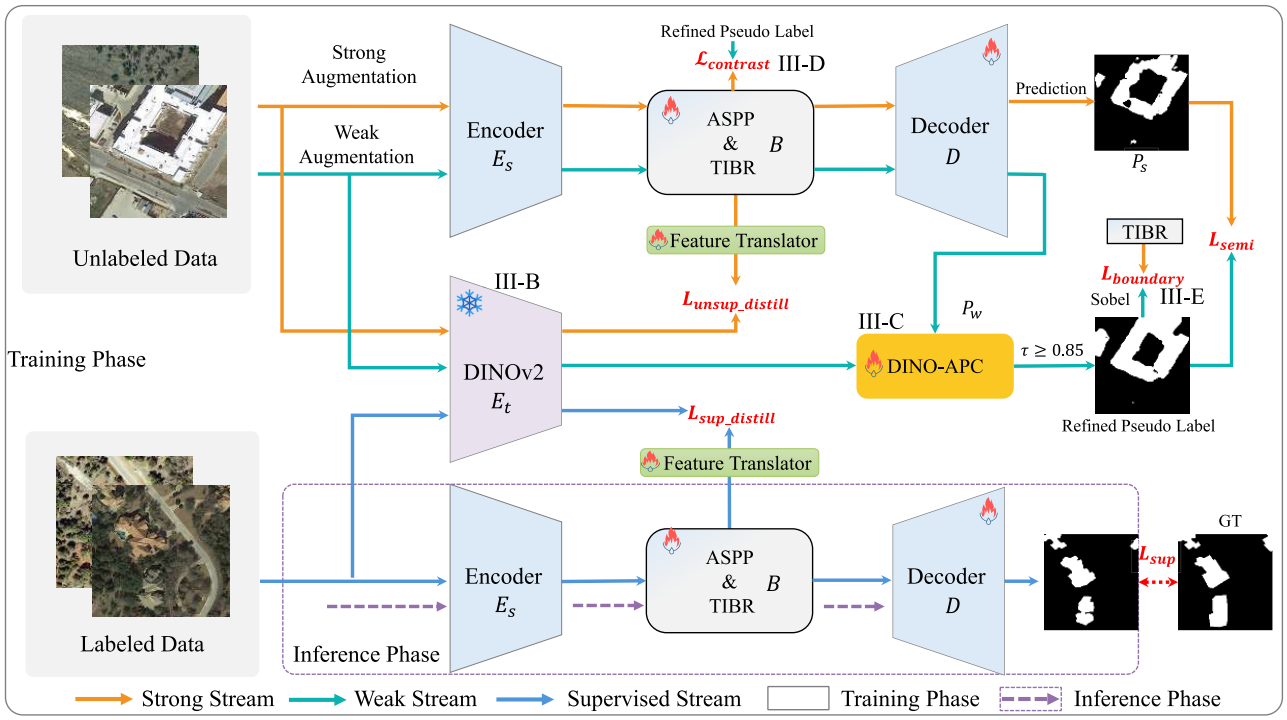


Fig. 1. Framework operates in two decoupled phases. During the training phase, unlabeled data is processed via the strong stream (orange) and weak stream (teal), producing predictions P_s and P_w, while labeled data is handled by the supervised stream (blue). A frozen DINOv2 teacher E_t (see Section III-B) transfers semantic knowledge to the student encoder E_s via feature translators. The DINO-APC module (see Section III-C) refines P_w into calibrated pseudo labels at threshold $\tau \geq 0.85$ to compute $\mathcal{L}_{semi}$. $\mathcal{L}_{contrast}$ (see Section III-D) and $\mathcal{L}_{boundary}$ (see Section III-E) further enhance feature discriminability and boundary precision. For visual clarity, auxiliary losses on the supervised branch are omitted but included in the full objective [see (27)–(29)]. During the inference phase, only the student branch $E_s \rightarrow B \rightarrow D$ is retained, achieving foundation-model-level performance at ResNet-level efficiency.

C. DINO-Based Adaptive Pseudolabel Calibration

In semisupervised training, pseudolabels generated from weakly augmented unlabeled data inevitably suffer from inherent noise. To address this, we propose a DINO-based adaptive pseudolabel calibration (DINO-APC) module that leverages DINOv2 features to refine pseudolabel confidence, thereby suppressing noisy predictions while reinforcing reliable ones. As illustrated in Fig. 2, the refinement strategy operates through a three-stage pipeline combining global change intensity estimation with local spatial response analysis to achieve robust confidence calibration.

Stage 1 Global Change Intensity Estimation: First, a lightweight change intensity estimator G is employed to assess the overall semantic change magnitude for each image pair

$$\begin{aligned} S_{\text{global}} &= G\left(\mathbf{F}_t^{\text{diff}}\right) \\ &= \sigma\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Conv}_{1\times1}\left(\text{GAP}\left(\mathbf{F}_t^{\text{diff}}\right)\right)\right)\right)\right) \end{aligned} \tag{7}$$

where GAP denotes global average pooling, which spatially aggregates F_t^{diff} into a compact global descriptor, and $\sigma(\cdot)$ denotes the sigmoid activation function. The resulting scalar $S_{\text{global}} \in [0, 1]$ indicates whether the image pair contains significant changes, providing a global context prior for subsequent confidence scaling.

Stage 2 Semantic Change Intensity Map: To capture pixel-wise semantic change patterns, a normalized semantic change intensity map is computed from the DINO change features. The L_2-norm along the channel dimension is first calculated to aggregate multiscale semantic information into a single-channel intensity map, which is then normalized to obtain spatially coherent change intensity scores

$$\mathbf{R}_{\text{spatial}} = \frac{\left\|\mathbf{F}_t^{\text{diff}}\right\|_2 - \min\left(\left\|\mathbf{F}_t^{\text{diff}}\right\|_2\right)}{\max\left(\left\|\mathbf{F}_t^{\text{diff}}\right\|_2\right) - \min\left(\left\|\mathbf{F}_t^{\text{diff}}\right\|_2\right) + \epsilon} \tag{8}$$

where $\|\cdot\|_2$ denotes the L_2-norm along channels, and $\epsilon = 10^{-8}$ ensures numerical stability. The normalized map $\mathbf{R}_{\text{spatial}} \in [0, 1]^{H\times W}$ highlights regions with strong semantic discrepancies, with higher values indicating more significant changes detected by the DINO pretrained representations.

Stage 3 Bidirectional Confidence Adjustment: Given the pseudolabels $\hat{Y}$ and the initial confidence map P_w from the weakly augmented student prediction, a two-step adaptive refinement process is performed, explicitly accounting for changed and unchanged regions. First, a bidirectional modulation map M_{dino} is constructed to align pseudolabel predictions with DINO's spatial response

$$M_{\text{dino}} = \begin{cases} \mathbf{R}_{\text{spatial}}, & \text{if } \hat{Y} = 1 \\ 1 - \mathbf{R}_{\text{spatial}}, & \text{if } \hat{Y} = 0. \end{cases} \tag{9}$$

The locally refined confidence is then computed as

$$P_{\text{local}} = \text{clamp}\left(P_w\left(1 + \alpha M_{\text{dino}}\right), 0, 1\right) \tag{10}$$

where $\text{clamp}(\cdot, 0, 1)$ clips values to the valid probability range $[0, 1]$, and $\alpha = 0.3$ controls the strength of spatial modulation.

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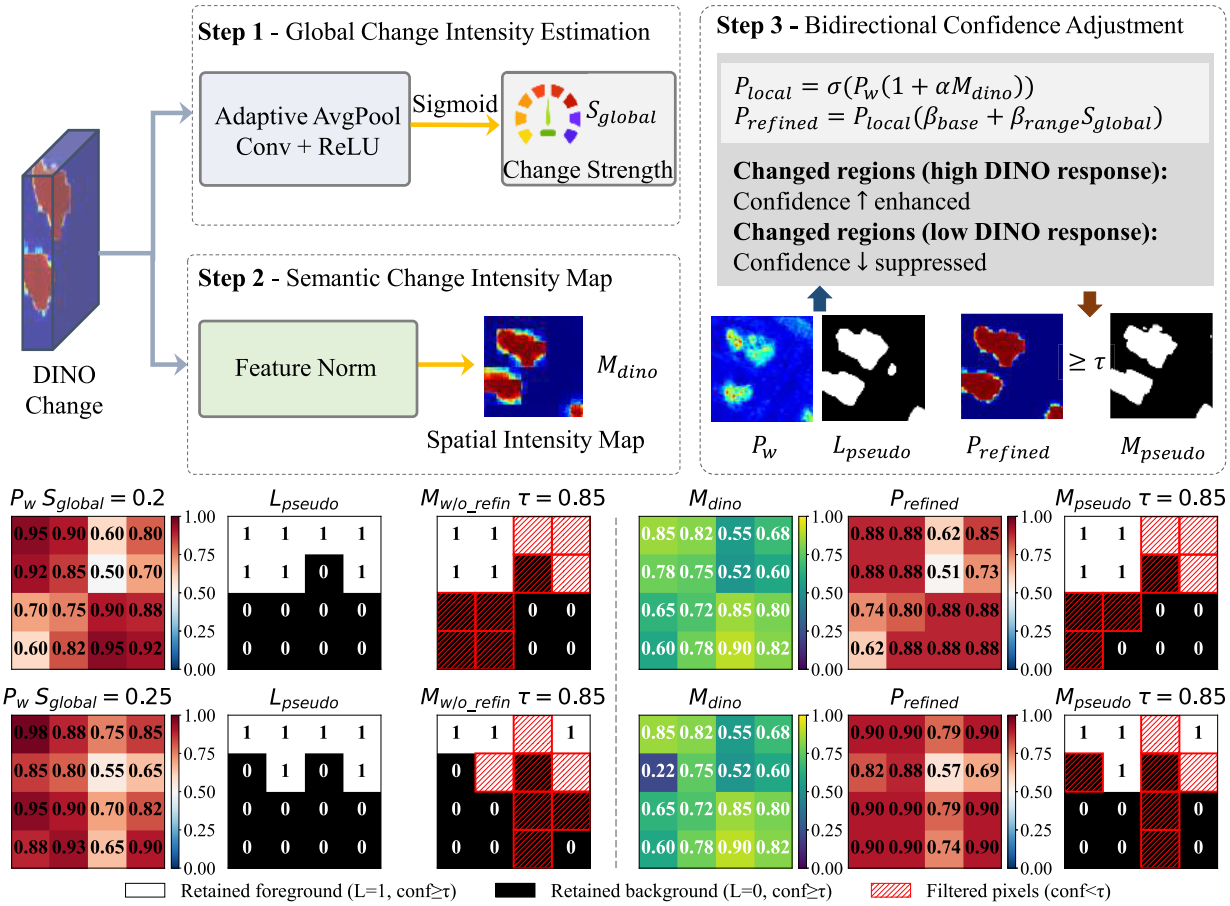


Fig. 2. Proposed DINO-APC module. It consists of: 1) global intensity estimation (S_{global}); 2) semantic change intensity map; and 3) bidirectional confidence adjustment. (Bottom) Evolution of the training mask for two samples. The initial prediction P_w is modulated by DINO-APC into P_{refined} and filtered ($\tau = 0.85$) to yield M_{pseudo}. The comparison with unrefined masks $M_{\text{w/o-refin}}$ highlights the suppression of noisy predictions (marked by red hatching), ensuring high-quality supervision.

Second, to account for variation in change magnitude across samples, a global scaling factor derived from S_{global} is applied

$$P_{\text{refined}} = P_{\text{local}} \cdot \left(\beta_{\text{base}} + \beta_{\text{range}} \cdot S_{\text{global}}\right) \tag{11}$$

where $\beta_{\text{base}} = 0.8$ and $\beta_{\text{range}} = 0.4$ are empirically set.

Finally, a training mask is generated by thresholding the refined confidence

$$M_{\text{pseudo}} = \mathbb{1}\left[P_{\text{refined}} \geq \tau\right] \tag{12}$$

where $\tau = 0.85$ is the confidence threshold and $\mathbb{1}[\cdot]$ is the indicator function. High-confidence pixels refer to those satisfying $P_{\text{refined}} \geq \tau$ ($\tau = 0.85$). They are retained by M_{pseudo} and used to compute the unsupervised loss. This formulation achieves the following effects (as illustrated in Fig. 2).

1) If the pseudolabel predicts "change" and DINO's spatial response is also high, confidence is boosted.
2) If the pseudolabel predicts "no change" and DINO's spatial response is also low, confidence is boosted.
3) Global change intensity S_{global} controls the overall confidence level.

In summary, the proposed refinement mechanism achieves a dual objective: it leverages DINO's semantic awareness to identify and suppress noisy predictions while preserving high-confidence regions that are consistent with the pretrained teacher's understanding of semantic change. As demonstrated in the bottom examples of Fig. 2, the method adaptively filters pseudolabels based on local spatial coherence and the global change intensity S_{global}.

D. Dual-Level Contrastive Learning

To augment feature discriminability and mitigate the propagation of pseudolabel noise, we propose a DLCL mechanism operating at two complementary levels: *intraimage* and *interbatch*, as illustrated in Fig. 3. Leveraging a synergistic local separation-global alignment constraint, this mechanism complements the aforementioned DINO-APC module. While the DINO-APC module focuses on filtering spatial noise within individual samples at the label input stage via confidence calibration, the DLCL mechanism addresses the residual noise that passes through this filtering stage by regulating intrasample feature topology and interbatch semantic consistency at the feature representation level. This two-stage collaboration forms a complementary label-feature dual-layer protection mechanism, wherein DINO-APC intercepts noise at the pseudolabel generation stage while DLCL suppresses its residual propagation at the feature representation stage.

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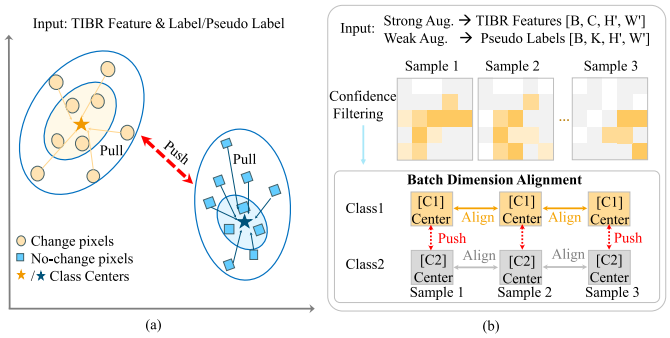


Fig. 3. Illustration of the DLCL mechanism. The mechanism comprises two complementary components. (a) Intraimage pixel-level contrastive learning ($\mathcal{L}_{intra}$): utilizes ground truth (for labeled data) or calibration masks (for unlabeled data) as gating signals to enforce separation between changed and unchanged features, enhancing local discriminability. (b) Batch-level prototype alignment ($\mathcal{L}_{inter}$): aligns class prototypes across samples within a batch using weighted centroids derived from calibration masks, correcting global distribution shift caused by pseudolabel noise.

1) Intraimage Contrastive Learning: Within each sample, we enforce feature separation between changed and unchanged regions. For a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ and label $\mathbf{Y}$, we first apply spatial L_2 normalization

$$\hat{\mathbf{F}}(i,j) = \frac{\mathbf{F}(:,i,j)}{\|\mathbf{F}(:,i,j)\|_2} \tag{13}$$

where $\hat{\mathbf{F}}(i,j) \in \mathbb{R}^C$ denotes the normalized feature vector at spatial position (i,j), and $\|\cdot\|_2$ represents the L_2 norm.

We then compute classwise prototypes by averaging features within each class

$$\boldsymbol{\mu}_c = \frac{1}{|\Omega_c|}\sum_{(i,j)\in\Omega_c} \hat{\mathbf{F}}(i,j), \quad c \in \{0,1\} \tag{14}$$

where $\Omega_c = \{(i,j) \mid \mathbf{Y}(i,j) = c\}$ denotes the pixel set of class c (0: unchanged, 1: changed), $|\Omega_c|$ is the cardinality of set Ω_c, and $\boldsymbol{\mu}_c$ represents the prototype (mean feature vector) of class c.

The intraimage loss pulls features toward their class prototype while pushing the two prototypes apart

$$\mathcal{L}_{\text{intra}} = \underbrace{\frac{1}{|\Omega_1|}\sum_{(i,j)\in\Omega_1}\left(1-\boldsymbol{\mu}_1^T\hat{\mathbf{F}}(i,j)\right)}_{\text{attraction term}} + \underbrace{\max\left(0,\boldsymbol{\mu}_1^T\boldsymbol{\mu}_0+m\right)}_{\text{repulsion term}} \tag{15}$$

where $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_0$ are prototypes for changed and unchanged regions respectively, $\boldsymbol{\mu}_1^T\hat{\mathbf{F}}(i,j)$ computes the cosine similarity between prototype and feature, and $m > 0$ is a margin hyperparameter encouraging interclass separation.

For different data types, this loss adopts differentiated anchor construction strategies.

1) *Labeled Data:* Directly constructs reliable feature anchors based on ground truth labels $\mathbf{Y}$, establishing the correct topology in feature space.
2) *Unlabeled Data:* Applies weighting through confidence mask $\mathbf{M}_{\text{pseudo}}$ (i.e., $\mathcal{L}_{\text{intra}} \cdot \mathbf{M}_{\text{pseudo}}$), imposing contrastive constraints only on high-confidence regions to avoid within each image pair misleading feature learning by noisy pseudolabels.

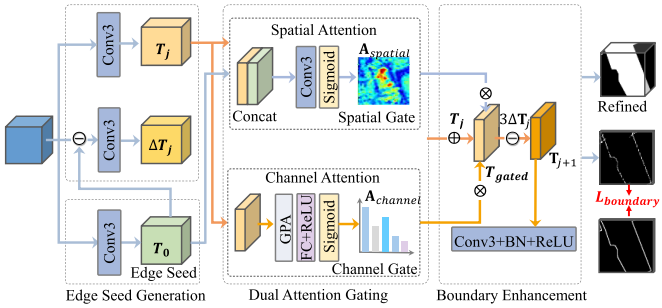


Fig. 4. Illustration of the TIBR module.

2) Batch-Level Prototype Alignment: Although intraimage contrast optimizes local feature structure, intraimage constraints cannot perceive cross-sample distribution consistency, easily leading to interbatch distribution drift induced by pseudolabel noise. To address this, we introduce a batch-level class center alignment loss to maintain semantic consistency at the global level.

For a batch of B samples with features $\{\mathbf{F}^b\}_{b=1}^B$, pseudolabels $\{\hat{\mathbf{Y}}^b\}_{b=1}^B$, and confidence masks $\{\mathbf{M}^b\}_{b=1}^B$ (where $\mathbf{M}^b(i,j) \in [0,1]$ indicates prediction confidence), we compute the weighted class center for each sample

$$\boldsymbol{\mu}_c^b = \frac{\sum_{(i,j)\in\Omega_c^b}\mathbf{M}^b(i,j)\cdot\hat{\mathbf{F}}^b(i,j)}{\sum_{(i,j)\in\Omega_c^b}\mathbf{M}^b(i,j)+\epsilon}, \quad c \in \{0,1\} \tag{16}$$

where $\Omega_c^b = \{(i,j) \mid \hat{\mathbf{Y}}^b(i,j) = c\}$ denotes the pixel set of class c in sample b, $\boldsymbol{\mu}_c^b$ is the prototype of class c for sample b, and ϵ is a small constant preventing division by zero. This weighting reduces the contribution of uncertain pseudolabels, biasing class centers toward high-confidence regions.

After L_2 normalization of the class centers, we compute pairwise similarities across the batch. The batch-level alignment loss encourages prototypes of the same class to remain consistent across different samples

$$\mathcal{L}_{\text{inter}} = \frac{1}{2BC^2}\sum_{c=0}^{1}\sum_{b_1=1}^{B}\sum_{b_2=1}^{B}\left(1-\frac{\left(\boldsymbol{\mu}_c^{b_1}\right)^T\boldsymbol{\mu}_c^{b_2}}{2}\right) \cdot \mathbb{1}\left[\boldsymbol{\mu}_c^{b_1}\neq\mathbf{0}\right]\cdot\mathbb{1}\left[\boldsymbol{\mu}_c^{b_2}\neq\mathbf{0}\right] \tag{17}$$

where $(\boldsymbol{\mu}_c^{b_1})^T\boldsymbol{\mu}_c^{b_2}$ measures the similarity between class c prototypes from samples b_1 and b_2, and $\mathbb{1}[\cdot]$ is the indicator function that excludes invalid centers (e.g., when class c is absent in a sample).

The proposed mechanism provides complementary local and global constraints. In particular, L_{intra} improves intraimage feature discrimination, whereas L_{inter} preserves intersample semantic consistency at the batch level. As a result, the model achieves more stable optimization and becomes less sensitive to local pseudolabel noise.

E. Taylor-Inspired Boundary Refinement

Low-confidence regions frequently concentrate near change boundaries, presenting a pervasive challenge for which existing methods lack dedicated optimization mechanisms. To address this limitation, we propose an iterative refinement

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process in feature space. The TIBR module operates on the ASPP output features $\mathbf{F}_{\text{ASPP}} \in \mathbb{R}^{C\times H\times W}$, as illustrated in Fig. 4. TIBR is Taylor-inspired in that it adopts the intuition of local corrective updating, but it is not a strict discretization or direct numerical implementation of the classical Taylor series in pixel space.

1) Edge Seed Generation: We first generate a coarse boundary estimation as initialization for iterative refinement

$$\mathbf{F}_{\text{seed}} = \text{BN}\left(\mathbf{W}_2^{\text{seed}} \cdot \text{ReLU}\left(\text{BN}\left(\mathbf{W}_1^{\text{seed}} \cdot \mathbf{F}_{\text{ASPP}}\right)\right)\right) \tag{18}$$

where $\mathbf{W}_1^{\text{seed}}, \mathbf{W}_2^{\text{seed}}$ are 3×3 convolutional weights.

2) Spatial Attention Gating: To localize uncertain boundary regions, we design a spatial attention mechanism that computes feature variance across spatial locations

$$\mathbf{A}_{\text{spatial}} = \sigma\left(\mathbf{W}_2^{\text{sp}} \cdot \text{ReLU}\left(\text{BN}\left(\mathbf{W}_1^{\text{sp}} \cdot \left[\mathbf{T}_j; \mathbf{F}_{\text{seed}}\right]\right)\right)\right) \tag{19}$$

where $[\cdot;\cdot]$ denotes channelwise concatenation, $\mathbf{T}_j$ represents the current boundary estimate at iteration j.

3) Channel Attention Gating: To select discriminative feature channels for boundary detection, we employ channel attention

$$\mathbf{A}_{\text{channel}} = \sigma\left(\mathbf{W}_2^{\text{ch}} \cdot \text{ReLU}\left(\mathbf{W}_1^{\text{ch}} \cdot \text{GAP}\left(\mathbf{T}_j\right)\right)\right) \tag{20}$$

where $\mathbf{A}_{\text{channel}} \in \mathbb{R}^{B\times 512\times 1\times 1}$ measures the importance of each feature channel for boundary localization.

4) Boundary Enhancement: Inspired by the first-order Taylor expansion, we formulate boundary refinement in feature space as a learnable iterative correction process, where each update applies a locally gated first-order adjustment to the current estimate. The boundary estimate is initialized at $j = 0$ as

$$\mathbf{T}_0 = \mathbf{W}_{\text{tj}}\left(\mathbf{F}_{\text{ASPP}}\right) \tag{21}$$

where $\mathbf{W}_{\text{tj}}$ is a 3×3 convolution. At each subsequent iteration $j \geq 0$, a discrepancy-guided correction signal is computed from the residual between the global context features and the coarse boundary seed

$$\Delta\mathbf{T}_j = \mathbf{W}_\delta\left(\mathbf{F}_{\text{ASPP}} - \mathbf{F}_{\text{seed}}\right) \tag{22}$$

where $\mathbf{W}_\delta$ is a 3×3 convolution. The correction is then selectively routed to uncertain boundary regions through dual attention gating

$$\mathbf{T}_{\text{gated}} = \Delta\mathbf{T}_j \odot \mathbf{A}_{\text{spatial}} \odot \mathbf{A}_{\text{channel}} \tag{23}$$

so that only high-uncertainty areas receive significant corrections while well-defined regions are left undisturbed. The boundary estimate is then updated as

$$\mathbf{T}_{j+1} = \mathbf{T}_j - \gamma\,\mathbf{T}_{\text{gated}} \tag{24}$$

where γ is a correction scaling coefficient controlling the update strength. In our implementation, γ is empirically set to 3 according to the sensitivity analysis.

The final refined features are obtained through boundary enhancement convolution and residual connection

$$\mathbf{F}_{\text{refined}} = \text{ReLU}\left(\text{BN}\left(\mathbf{W}_{\text{enh}}\left(\mathbf{T}_{j+1}\right)\right)\right) + \mathbf{F}_{\text{ASPP}}. \tag{25}$$

To explicitly supervise boundary learning, we extract edge maps $\mathbf{E}_{\text{gt}}$ from ground truth or pseudo labels using the Sobel operator, and supervise the spatial attention map with binary cross-entropy loss

$$\mathcal{L}_{\text{boundary}} = \text{BCE}\left(\mathbf{A}_{\text{spatial}}, \mathbf{E}_{\text{gt}}\right). \tag{26}$$

F. Overall Training Objective

The total loss $\mathcal{L}_{\text{total}}$ is composed of the labeled-branch objective $\mathcal{L}_l$ and the unlabeled-branch objective $\mathcal{L}_u$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_l + \mathcal{L}_u. \tag{27}$$

The labeled branch establishes the foundational mapping capabilities using explicit supervision from labeled data and auxiliary feature refinement, while the unlabeled branch ensures the global semantic stability of the model through consistency regularization, global distribution alignment, and auxiliary refinement on reliable pseudolabeled regions

$$\mathcal{L}_l = \mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{aux}}^l, \qquad \mathcal{L}_u = \mathcal{L}_{\text{semi}} + \mathcal{L}_{\text{inter}} + \mathcal{L}_{\text{aux}}^u. \tag{28}$$

In particular, $\mathcal{L}_{\text{sup}}$ applies standard cross-entropy on labeled data $\mathcal{D}_{\text{L}}$. For unlabeled data $\mathcal{D}_{\text{U}}$, $\mathcal{L}_{\text{semi}}$ enforces weak–strong augmentation consistency using pseudolabels filtered by the reliable mask M, where the consistency is measured by cross-entropy loss. Significantly, to mitigate the distribution drift often induced by noisy pseudolabels, we incorporate the batch-level alignment loss $\mathcal{L}_{\text{inter}}$ into the unlabeled branch as an inherent component of the core learning framework. As detailed in Section III-D, $\mathcal{L}_{\text{inter}}$ aligns class prototypes across batches, thereby maintaining global semantic consistency during training. This role distinguishes it from $\mathcal{L}_{\text{intra}}$, whose primary function is local feature refinement within individual samples and which is therefore more appropriately categorized as an auxiliary term.

The auxiliary terms focus on optimizing local feature topology and boundary precision. The auxiliary loss aggregates KD, boundary supervision, and intraimage contrastive learning

$$\mathcal{L}_{\text{aux}}^b = \lambda_1\mathcal{L}_{\text{distill}}^b + \lambda_2\mathcal{L}_{\text{boundary}}^b + \lambda_3\mathcal{L}_{\text{intra}}^b, \qquad b \in \{l, u\}. \tag{29}$$

Here, $\mathcal{L}_{\text{distill}}$ enforces semantic consistency between the teacher and student networks, while $\mathcal{L}_{\text{boundary}}$ sharpens prediction edges. The intraimage loss $\mathcal{L}_{\text{intra}}$ specifically targets local feature separability within individual samples. By pushing apart pixel embeddings of changed and unchanged regions, $\mathcal{L}_{\text{intra}}$ refines the decision boundaries in the feature space, complementing the global alignment provided by $\mathcal{L}_{\text{inter}}$.

Based on empirical validation in Section IV, the hyperparameters are set to $\lambda_1 = 0.3$, $\lambda_2 = 0.15$, and $\lambda_3 = 0.1$. The complete training and inference procedure of DistillSemiCD is summarized in Algorithm 1.

IV. Experiment and Results

A. Datasets

To comprehensively evaluate the effectiveness and generalization capability of the proposed method, experiments were conducted on four mainstream RSCD benchmark datasets: WHU-CD [64], SYSU-CD [65], CDD-CD [66], and LEVIR-CD [67]. An SSL setting was adopted, where subsets of the training set (5%, 10%, 20%, and 40%) were randomly

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Algorithm 1 Overall Framework of DistillSemiCD

Require: labeled set D_l, unlabeled set D_u, student encoder E_s, boundary refinement module B, decoder D, frozen teacher E_t, feature translator T, confidence threshold τ

for each training iteration **do**
  Sample labeled data (I_A^l, I_B^l, Y_l)
  Sample unlabeled weak and strong views (W_A^u, W_B^u) and (S_A^u, S_B^u)
  Labeled branch:
  $F_l \leftarrow E_s(I_A^l, I_B^l)$
  $P_l \leftarrow D(B(F_l))$
  Compute supervised loss $\mathcal{L}_{sup}$ using P_l and Y_l
  $F_t^l \leftarrow E_t(I_A^l, I_B^l)$
  Compute labeled auxiliary loss $\mathcal{L}_{aux}^l$, including distillation, boundary, and intraimage contrastive losses
  Unlabeled branch:
  $P_w \leftarrow D(B(E_s(W_A^u, W_B^u)))$
  Generate pseudo label $\hat{Y}_u$ and confidence map C_u from P_w
  Refine C_u with the teacher prior and obtain refined confidence map C_r
  $M \leftarrow \mathbf{1}(C_r \geq \tau)$
  $P_s \leftarrow D(B(E_s(S_A^u, S_B^u)))$
  Compute masked unsupervised loss $\mathcal{L}_{semi}$ using P_s, $\hat{Y}_u$, and M
  Compute interimage contrastive loss $\mathcal{L}_{inter}$ on reliable unlabeled regions
  Compute unlabeled auxiliary loss $\mathcal{L}_{aux}^u$ on reliable unlabeled regions
  $\mathcal{L} \leftarrow (\mathcal{L}_{sup} + \mathcal{L}_{aux}^l) + (\mathcal{L}_{semi} + \mathcal{L}_{inter} + \mathcal{L}_{aux}^u)$
  Update the student network parameters while keeping E_t frozen
end for
  Inference:
$P \leftarrow D(B(E_s(I_A, I_B)))$
return P

selected as labeled data, while the remaining samples served as unlabeled data.

1) *WHU-CD:* This dataset consists of a single pair of high-resolution aerial images with a spatial resolution of 0.2 m/pixel. It focuses on building morphological changes. The original imagery was cropped into nonoverlapping 256 × 256 patches and split into 5947/744/743 samples for training, validation, and testing.
2) *SYSU-CD:* This dataset contains 20 000 image pairs sourced from satellite imagery with a spatial resolution of 0.5 m/pixel. It covers diverse object changes, including buildings, ships, roads, and vegetation. The dataset is partitioned into 12 000/4000/4000 samples for training, validation, and testing.
3) *CDD-CD:* This dataset comprises 16 000 multiseasonal image pairs sourced from Google Earth with a wide variation in spatial resolution (0.03–1 m/pixel). It focuses on diverse change types such as phenological vegetation changes, urban expansion, and land use conversion. The dataset is divided into 10 000/3000/3000 samples for training, validation, and testing.
4) *LEVIR-CD:* This dataset contains high-resolution aerial images sourced from Google Earth with a spatial resolution of 0.5 m/pixel, spanning 2002 to 2018. It focuses primarily on urban building changes, including construction and demolition. The dataset is split into 7120/1024/2048 samples for training, validation, and testing.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: The proposed method was implemented using PyTorch 2.4 framework with Python 3.10 and executed on a server equipped with an NVIDIA GeForce RTX 4090 GPU. The network was trained using the Adam optimizer with an initial learning rate of 1×10^{-4}, a batch size of 8. Data augmentation was divided into weak and strong strategies. Weak augmentation was used to generate stable pseudolabel references and included random horizontal flipping, random rescaling within the range of $[0.5, 2.0]$, and random cropping to the input size. Strong augmentation followed the RandAugment strategy in FPA [51], where two operations were randomly sampled from nine photometrically meaningful transformations: identity, contrast, AutoContrast, brightness, color, equalize, sharpness, posterize, and solarize.

2) Evaluation Metrics: To assess the performance, we employed $F1$-score and intersection over union (IoU) as evaluation metrics. The $F1$-score is the harmonic mean of precision and recall, providing a more balanced evaluation between false alarms and missed detections. These metrics are defined as

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{30}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{31}$$

where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative pixels, respectively.

C. Comparison With State-of-the-Art Methods

Table I presents the quantitative comparison between the proposed DistillSemiCD and representative semisupervised change detection methods. In addition to conventional semisupervised baselines, including S4GAN [30], SemiCDNet [42], SemiCD [43], RCL [50], UniMatch [33], FPA [51], ECPS [47], CutMix-CD [44], and AdaSemiCD [48], we further include recent VFM/VLM-related methods, namely SemiCD-VL [53], SemiBCD [54], and UniMatch-V2 [29], on the WHU-CD and LEVIR-CD datasets, since these methods were originally validated primarily on these two benchmarks and we reproduce them under identical experimental settings for a focused and fair comparison. The 100% labeled results are also reported as the full-supervision upper bound of DistillSemiCD, enabling a clearer assessment of the gap between SSL and fully supervised training.

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TABLE I

COMPREHENSIVE QUANTITATIVE COMPARISON ON WHU-CD, SYSU-CD, CDD-CD, AND LEVIR-CD DATASETS. RED, BLUE, AND BOLD INDICATE THE BEST, SECOND-BEST, AND THIRD-BEST METRICS, RESPECTIVELY

	No.	Methods	Pub.	5%		10%		20%		40%	
				IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)
WHU-CD	1	S4GAN [30]	TPAMI19	46.24	63.24	61.92	76.48	70.71	82.84	76.43	86.64
	2	SemiCDNet [42]	TGRS20	53.28	69.52	61.13	75.88	66.59	79.94	75.13	85.80
	3	SemiCD [43]	Arxiv22	65.80	79.37	68.17	81.07	74.82	85.60	77.26	87.17
	4	RCL [50]	TGRS22	69.70	82.14	71.88	83.64	75.31	85.92	78.44	87.92
	5	UniMatch [33]	CVPR23	76.38	86.61	78.32	87.84	80.36	89.11	84.27	91.46
	6	FPA [51]	TGRS23	70.85	82.94	75.33	85.93	77.16	87.11	80.60	89.26
	7	ECPS [47]	TGRS24	43.19	60.33	43.93	61.04	50.85	67.42	55.83	71.66
	8	CutMix-CD [44]	TGRS25	74.49	85.38	78.88	88.19	81.25	89.66	86.47	92.74
	9	AdaSemiCD [48]	TGRS25	67.80	80.81	70.80	82.90	74.70	85.52	79.60	88.64
	10	SemiCD-VL [53]	TGRS24	**81.27**	**89.67**	82.76	90.57	83.96	91.28	86.70	92.88
	11	SemiBCD [54]	TGRS25	**81.70**	**89.93**	**84.08**	**91.35**	**84.23**	**91.44**	**88.01**	**93.62**
	12	UniMatch-V2 [29]	TPAMI25	**81.36**	**89.72**	**84.78**	**91.76**	**84.49**	**91.59**	**88.74**	**94.03**
	13	**DistillSemiCD**	**—**	80.40	89.14	**84.56**	**91.63**	**84.38**	**91.53**	**87.87**	**93.54**
	—	100% label	—	—	—	—	—	—	—	88.98	94.17
SYSU-CD	1	S4GAN [30]	TPAMI19	52.73	69.05	57.03	72.64	60.56	75.44	61.84	76.42
	2	SemiCDNet [42]	TGRS20	49.86	66.54	50.63	67.22	54.87	70.86	61.07	75.83
	3	SemiCD [43]	Arxiv22	50.61	67.21	51.68	68.14	56.08	71.86	64.21	78.20
	4	RCL [50]	TGRS22	57.91	73.35	60.65	75.51	62.86	77.20	64.45	78.38
	5	UniMatch [33]	CVPR23	**66.21**	**79.67**	**67.51**	**80.60**	68.19	81.09	69.23	81.82
	6	FPA [51]	TGRS23	64.63	78.52	66.41	79.81	68.13	81.04	68.83	81.54
	7	ECPS [47]	TGRS24	57.29	72.85	62.28	76.76	63.34	77.56	63.77	77.88
	8	CutMix-CD [44]	TGRS25	62.61	77.01	66.75	80.06	**68.98**	**81.64**	**69.46**	**81.98**
	9	AdaSemiCD [48]	TGRS25	**67.50**	**80.60**	**68.70**	**81.45**	**70.10**	**82.42**	**69.90**	**82.28**
	10	**DistillSemiCD**	**—**	**70.75**	**82.87**	**71.25**	**83.21**	**71.82**	**83.60**	**71.98**	**83.71**
	—	100% label	—	—	—	—	—	—	—	72.81	84.27
CDD-CD	1	S4GAN [30]	TPAMI19	62.37	76.82	71.02	83.05	79.06	88.31	82.81	90.60
	2	SemiCDNet [42]	TGRS20	63.52	77.69	71.29	83.24	79.14	88.36	82.83	90.61
	3	SemiCD [43]	Arxiv22	67.65	80.70	75.55	86.07	80.23	89.03	82.74	90.55
	4	RCL [50]	TGRS22	68.10	81.02	76.51	86.69	80.73	89.34	82.24	90.25
	5	UniMatch [33]	CVPR23	**68.63**	**81.40**	74.55	85.42	**83.78**	**91.17**	84.23	91.44
	6	FPA [51]	TGRS23	66.59	79.94	74.49	85.38	81.02	89.51	83.07	90.75
	7	ECPS [47]	TGRS24	68.08	81.01	**78.25**	**87.80**	**85.42**	**92.14**	**86.66**	**92.85**
	8	CutMix-CD [44]	TGRS25	67.70	80.74	74.33	85.28	80.93	89.46	**86.73**	**92.89**
	9	AdaSemiCD [48]	TGRS25	**70.10**	**82.42**	**77.30**	**87.20**	82.10	90.17	83.90	91.25
	10	**DistillSemiCD**	**—**	**72.08**	**83.77**	**79.41**	**88.52**	**87.94**	**93.58**	**90.82**	**95.19**
	—	100% label	—	—	—	—	—	—	—	92.83	96.28
LEVIR-CD	1	S4GAN [30]	TPAMI19	66.53	79.90	71.30	83.25	73.98	85.04	76.02	86.38
	2	SemiCDNet [42]	TGRS20	69.42	81.95	73.57	84.77	75.43	85.99	76.36	86.60
	3	SemiCD [43]	Arxiv22	74.19	85.18	77.12	87.08	77.85	87.55	78.99	88.26
	4	RCL [50]	TGRS22	69.29	81.86	73.12	84.47	73.52	84.74	74.61	85.46
	5	UniMatch [33]	CVPR23	78.32	87.84	79.88	88.81	80.11	88.96	80.31	89.08
	6	FPA [51]	TGRS23	73.47	84.71	75.63	86.12	77.35	87.23	78.51	87.96
	7	ECPS [47]	TGRS24	75.43	85.99	78.39	87.89	79.43	88.54	80.42	89.15
	8	CutMix-CD [44]	TGRS25	72.21	83.86	77.63	87.41	79.68	88.69	80.75	89.35
	9	AdaSemiCD [48]	TGRS25	77.70	87.45	79.40	88.52	80.30	89.07	80.60	89.26
	10	SemiCD-VL [53]	TGRS24	**80.58**	**89.25**	81.83	90.01	82.16	90.21	**82.28**	**90.28**
	11	SemiBCD [54]	TGRS25	**81.62**	**89.88**	**82.48**	**90.40**	**82.74**	**90.55**	**83.03**	**90.73**
	12	UniMatch-V2 [29]	TPAMI25	**81.75**	**89.96**	**82.51**	**90.42**	**82.72**	**90.54**	**83.58**	**91.06**
	13	**DistillSemiCD**	**—**	79.68	88.69	**82.06**	**90.15**	**82.44**	**90.37**	82.00	90.11
	—	100% label	—	—	—	—	—	—	—	83.44	90.97

1) Quantitative Evaluation: Table I summarizes the comprehensive results across four benchmarks under different labeled-data ratios. Overall, DistillSemiCD achieves competitive or top-tier IoU/$F1$ performance while maintaining

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a moderate inference cost. After incorporating stronger VFM/VLM-related baselines, our method is no longer the best in every WHU-CD and LEVIR-CD setting, but it still provides a favorable balance between accuracy and efficiency. This competitiveness mainly stems from the proposed training–inference decoupled distillation strategy, which transfers structural priors from a frozen DINOv2 teacher to the lightweight student during training and removes the teacher during inference.

On the WHU-CD dataset, DistillSemiCD achieves 80.40%/89.14%, 84.56%/91.63%, 84.38%/91.53%, and 87.87%/93.54% IoU/$F1$ under the 5%, 10%, 20%, and 40% labeled ratios, respectively. It ranks second under the 10% and 20% settings, with only 0.22% and 0.11% IoU gaps compared with UniMatch-V2. At the 40% ratio, DistillSemiCD remains competitive with recent VFM/VLM-based methods and reduces the gap to the 100% labeled upper bound to 1.11% IoU and 0.63% $F1$.

For the SYSU-CD dataset, DistillSemiCD consistently achieves the best results across all labeled ratios. It obtains 70.75%/82.87%, 71.25%/83.21%, 71.82%/83.60%, and 71.98%/83.71% IoU/$F1$ from 5% to 40% labels, outperforming all compared methods. Since SYSU-CD contains heterogeneous changes involving buildings, roads, vegetation, and other land-cover categories, these results indicate that DINO-APC and DLCL effectively suppress pseudolabel noise and stabilize change-aware feature representations. Compared with the 100% labeled upper bound, the 40% setting leaves only a 0.83% IoU and 0.56% $F1$ gap.

On the CDD-CD dataset, DistillSemiCD also shows clear advantages over existing semisupervised baselines. Its performance improves from 72.08%/83.77% IoU/$F1$ at the 5% labeled ratio to 90.82%/95.19% at the 40% labeled ratio, demonstrating strong scalability with additional labeled data. At 40%, it exceeds the second-best method by 4.09% IoU and 2.30% $F1$. These results verify the effectiveness of the proposed distillation and contrastive alignment mechanisms under seasonal variations, resolution differences, and severe class imbalance.

On the LEVIR-CD dataset, VFM/VLM-related methods further strengthen the benchmark. DistillSemiCD achieves 79.68%/88.69%, 82.06%/90.15%, 82.44%/90.37%, and 82.00%/90.11% IoU/$F1$ under the 5%, 10%, 20%, and 40% labeled ratios, respectively. Although SemiBCD and UniMatch-V2 obtain slightly higher scores in several settings, DistillSemiCD remains close to them, especially under the 10% and 20% ratios. The 40% result is only 1.44% IoU and 0.86% $F1$ lower than the 100% labeled upper bound, showing that our method can approach full-supervision performance without introducing a heavy foundation-model detector at inference.

2) Qualitative Evaluation: Fig. 5 presents the visual comparisons across different datasets and labeled ratios. From an overall visual perspective, the detection maps produced by DistillSemiCD are visually competitive and cleaner in the selected examples, exhibiting minimal red and green regions, which indicates that the model effectively suppresses false positives while maintaining high recall.

In particular, on the WHU-CD dataset with an extremely low labeled ratio of 5%, existing methods suffer from severe boundary blurring and structural omission (extensive green areas). On the contrary, our method, leveraging the TIBR module, generates prediction maps with crisp contours and complete building structures. On the complex SYSU-CD and CDD-CD datasets, where vegetation interference and seasonal changes are prevalent, competing methods tend to produce fragmented false positives (red noise). However, our method demonstrates superior noise immunity and robustness, attributed to the strong semantic modeling capabilities of DINO-APC and DLCL. Furthermore, on LEVIR-CD (40%), our approach confirms its superiority in dense building change detection by accurately reconstructing fine-grained object details.

3) Qualitative Analysis on Challenging Samples: Fig. 6 presents qualitative comparisons on representative challenging samples from LEVIR-CD (rows 1–2) and WHU-CD (rows 3–4) under the 10% labeled setting, with VFM/VLM-based methods included for reference. In Row 1, small building changes are affected by illumination variation and low color contrast. DINO-APC provides more reliable change-confidence cues in such ambiguous regions, leading to a more complete and spatially coherent prediction. In Row 2, some changed buildings are partially absent from the reference annotation. Although these detections may be penalized as false positives, DLCL helps capture semantically consistent change cues supported by the bitemporal images. In Row 3, a probable annotation inconsistency is observed, where a visually plausible change is not fully reflected in the ground truth. DistillSemiCD still identifies this region, suggesting the usefulness of DINO-APC under imperfect supervision. In Row 4, the changed object has a circular structure requiring shape-level discrimination and accurate boundary localization. TIBR produces a more compact and geometrically consistent prediction through iterative discrepancy-guided refinement. Overall, these results show that DINO-APC, DLCL, and TIBR improve robustness under weak visual contrast, imperfect annotations, and complex object boundaries.

D. Ablation Studies

To systematically validate the efficacy of the proposed modules and their synergistic effects, comprehensive ablation studies were conducted on four change detection datasets under a 10% labeled data setting. Table II presents the performance comparison upon progressively removing core components from the full model. In particular, the baseline model is composed only of ResNet50, ASPP, and a lightweight decoder, without incorporating any of the proposed modules, namely TID-KD, DINO-APC, DLCL, and TIBR. The experimental results clearly reveal the individual contribution of each module and the overall collaborative advantage of the framework.

1) Effectiveness of TID-KD: Removing the TID-KD module resulted in significant performance degradation. In particular, the IoU on the WHU-CD dataset dropped by 1.95%, while the complex CDD-CD dataset witnessed the largest decline

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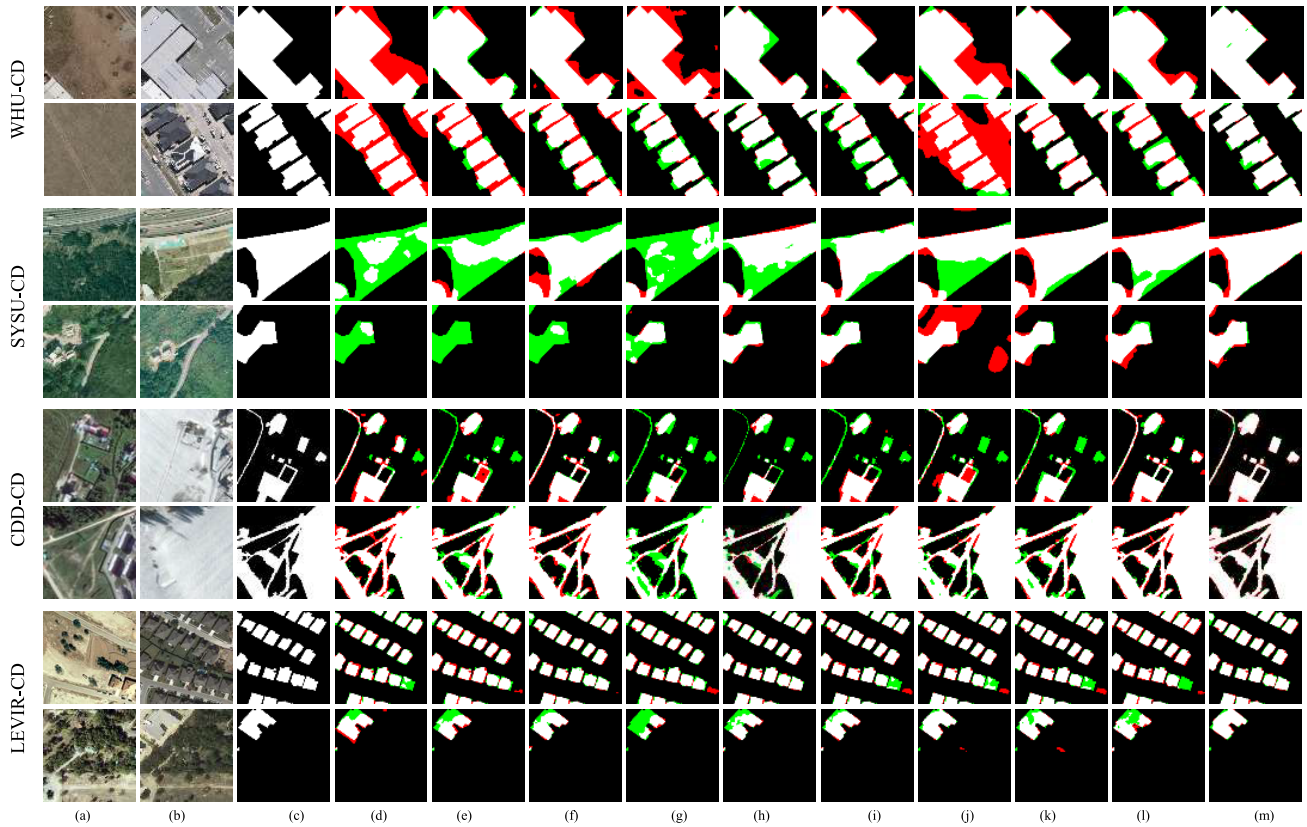


Fig. 5. Visualization comparison of results on four datasets under different labeled data ratios, including WHU-CD at 5%, SYSU-CD at 10%, CDD-CD at 20%, and LEVIR-CD at 40% labeled data. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (green). The less red and green, the more accurate the detection. (a) Preimage. (b) Postimage. (c) GT. (d) S4GAN. (e) SemiCDNet. (f) SemiCD. (g) RCL. (h) UniMatch. (i) FPA. (j) ECPS. (k) CutMix-CD. (l) AdaSemiCD. (m) Ours.

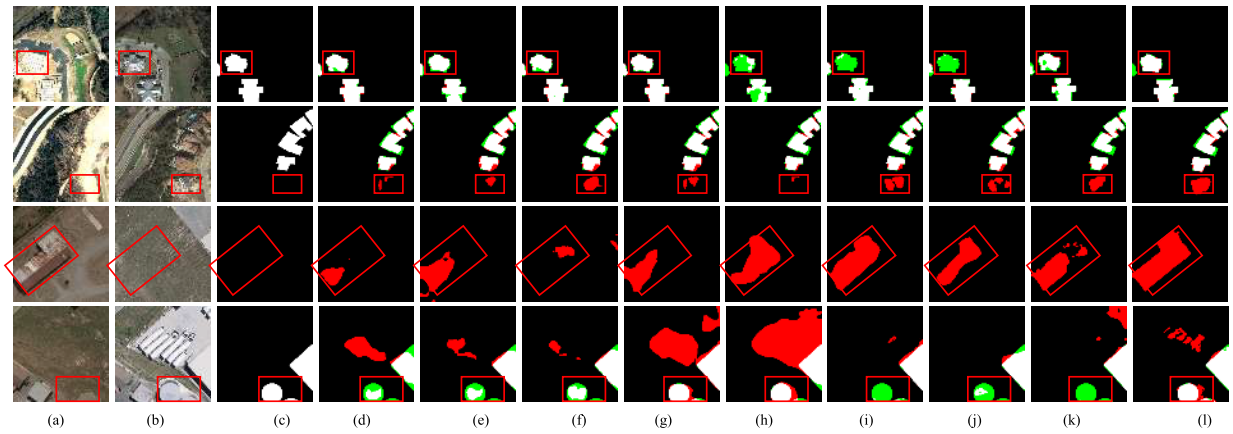


Fig. 6. Qualitative comparison on representative challenging samples from LEVIR-CD (rows 1–2) and WHU-CD (rows 3–4). (a) Preimage. (b) Postimage. (c) GT. (d) UniMatch. (e) FPA. (f) ECPS. (g) CutMix-CD. (h) AdaSemiCD. (i) SemiCD-VL. (j) SemiBCD. (k) Unimatch-V2. (l) Ours.

of 2.58%. This indicates that under label-scarce conditions, transferring rich semantic priors from the DINOv2 foundation model to the student network is critical. It effectively mitigates overfitting and insufficient generalization issues caused by the limited data available to the encoder. It is worth noting that the "w/o TID-KD" variant directly ablates the KD component, degrading to a student network trained without any distillation guidance, which further confirms that distillation is indispensable for performance in label-scarce settings.

2) Effectiveness of DINO-APC: The exclusion of the DINO-APC module led to IoU decreases of 1.22%–1.06% on the WHU-CD and CDD-CD datasets, respectively. By leveraging DINO's semantic change intensity maps to adaptively adjust confidence scores, this module suppresses the generation of low-quality pseudolabels at the source. Consequently, it contributes significantly to training stability, particularly in small-sample and complex change scenarios. Nevertheless, DINOv2 may have limited sensitivity to some fine-grained

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TABLE II
ABLATION STUDIES ON FOUR DATASETS UNDER 10% LABELED RATIO. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	WHU-CD IoU/F1	SYSU-CD IoU/F1	CDD-CD IoU/F1	LEVIR-CD IoU/F1
Baseline	81.74/89.95	69.25/81.83	77.16/87.11	80.49/89.19
w/o TID-KD	82.61/90.48	70.59/82.76	76.83/86.90	81.27/89.67
w/o DINO-APC	83.34/90.91	70.98/83.03	78.35/87.86	81.90/90.05
w/o DLCL	82.82/90.60	70.41/82.64	79.35/88.49	81.84/90.01
w/o TIBR	82.69/90.52	70.66/82.81	77.76/87.49	81.74/89.95
w/ DINOv3	83.67/91.11	70.69/82.83	78.21/87.77	81.29/89.68
Full Model	**84.56/91.63**	**71.25/83.21**	**79.41/88.52**	**82.06/90.15**

TABLE III
ABLATION STUDY ON CONFIDENCE THRESHOLD τ ON WHU-CD AND SYSU-CD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

τ	WHU-CD 5% IoU/F1	WHU-CD 10% IoU/F1	SYSU-CD 5% IoU/F1	SYSU-CD 10% IoU/F1
0.75	80.61/89.26	83.51/91.01	70.16/82.46	69.64/82.10
0.80	80.54/89.22	83.70/91.13	70.45/82.66	70.72/82.85
0.85	80.40/89.14	**84.56/91.63**	**70.75/82.87**	**71.25/83.21**
0.90	**81.16/89.60**	83.33/90.91	70.57/82.75	70.78/82.89
0.95	80.73/89.34	82.72/90.54	69.48/81.99	70.90/82.97

TABLE IV
PARAMETER SENSITIVITY ANALYSIS OF LOSS WEIGHTS $(\lambda_1, \lambda_2, \lambda_3)$ UNDER 5% LABELED RATIO

Param.	Value	WHU-CD IoU/F1	SYSU-CD IoU/F1	CDD-CD IoU/F1	LEVIR-CD IoU/F1
λ_1	0.1	**80.93/89.46**	70.81/82.91	69.72/82.16	78.62/88.03
	0.2	80.15/88.98	**71.10/83.11**	70.41/82.64	78.89/88.20
	0.3	80.40/89.14	70.75/82.87	**72.08/83.77**	**79.68/88.69**
	0.5	80.23/89.03	71.08/83.10	71.87/83.63	79.63/88.66
λ_2	0.1	80.18/89.00	70.67/82.81	72.01/83.73	79.07/88.31
	0.15	80.40/89.14	**70.75/82.87**	**72.08/83.77**	**79.68/88.69**
	0.2	**81.01/89.51**	70.53/82.72	70.87/82.95	78.80/88.14
λ_3	**0.1**	80.40/89.14	**70.75/82.87**	**72.08/83.77**	**79.68/88.69**
	0.15	**80.71/89.33**	70.16/82.46	70.14/82.45	79.35/88.49
	0.2	79.66/88.68	70.56/82.74	70.26/82.53	79.64/88.67

remote sensing changes due to its natural-image pretraining. Since DINO-APC only uses DINOv2 responses as auxiliary confidence priors, remote sensing foundation models could be incorporated as alternative teachers in future work.

3) Effectiveness of DLCL: On the WHU-CD and SYSU-CD datasets, the removal of the DLCL mechanism caused IoU drops of 1.74% and 0.84%, respectively. This mechanism enhances feature discrimination by enforcing intraclass feature aggregation and interclass separation. Notably, its contribution to the CDD-CD dataset, which contains substantial seasonal changes, was relatively minor (0.06%), suggesting that the gains from this module are closely correlated with the inherent feature separability of the scene.

4) Effectiveness of TIBR: The impact of removing the TIBR module was particularly pronounced on boundary-intensive datasets (e.g., a 1.87% IoU drop on WHU-CD). Utilizing Taylor finite difference iterations coupled with dual attention gating, TIBR specifically targets boundary blurring caused by down-sampling operations, effectively improving the localization accuracy of boundaries in changed regions.

5) Comparison Between DINOv2 and DINOv3: We replaced the frozen DINOv2 teacher with DINOv3 under the same TID-KD framework. As illustrated in Table II, the DINOv3-based variant consistently yields lower IoU scores across all four datasets, suggesting that DINOv2's 11th Transformer layer offers a more favorable balance between semantic abstraction and spatial localization for bitemporal change detection.

E. Hyperparameter Analysis

1) Impact of Confidence Threshold τ: Table III presents the sensitivity analysis of the confidence threshold τ for pseudolabel mask generation. The results indicate that $\tau = 0.85$ provides a robust overall tradeoff among the reported WHU-CD and SYSU-CD settings. A lower threshold of $\tau = 0.75$ admits residual noisy predictions despite DINO-APC suppression, reducing the SYSU-CD IoU to 70.16% at the 5% labeling ratio. An overly strict threshold of $\tau = 0.95$ aggressively filters high-confidence but moderate-scoring samples in the range $0.85 \le P_{\text{refined}} < 0.95$, degrading the SYSU-CD IoU to 69.48%. The selected value $\tau = 0.85$ better aligns with the confidence distribution of P_{refined}, yielding consistent stability across datasets and labeling ratios.

2) Parameter Sensitivity Analysis: Table IV reports the sensitivity of the loss weights λ_1 for distillation, λ_2 for boundary learning, and λ_3 for contrastive learning. The configuration $\{\lambda_1 = 0.3, \lambda_2 = 0.15, \lambda_3 = 0.1\}$ yields the most robust overall performance.

1) *Distillation Weight λ_1:* Complex scenes, such as CDD-CD and LEVIR-CD, require sufficient distillation guidance to learn structural representations, while larger values may introduce overconstraint. We select $\lambda_1 = 0.3$ as a balanced choice.
2) *Boundary Weight λ_2:* A value of 0.15 provides stable gains across benchmarks. Although a higher weight, e.g., $\lambda_2 = 0.2$, benefits WHU-CD due to its distinct building edges, it degrades SYSU-CD and LEVIR-CD by amplifying high-frequency noise.
3) *Contrastive Weight λ_3:* A value of 0.1 preserves the auxiliary role of contrastive learning, enhancing feature discriminability without overshadowing the primary segmentation objective on complex datasets.

3) Sensitivity Analysis of Margin m: As illustrated in Table V, the proposed method maintains stable performance

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TABLE V
SENSITIVITY ANALYSIS OF THE MARGIN m ON WHU-CD AND SYSU-CD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

m	WHU-CD		SYSU-CD	
	5%	10%	5%	10%
	IoU/F1	IoU/F1	IoU/F1	IoU/F1
0.1	80.24/89.04	83.64/91.09	70.17/82.47	70.21/82.50
0.2	**80.48/89.18**	83.42/90.96	70.14/82.45	70.38/82.62
0.3	80.40/89.14	**84.56/91.63**	**70.75/82.87**	**71.25/83.21**
0.4	80.36/89.11	83.51/91.01	70.35/82.59	70.71/82.84
0.5	79.90/88.83	84.05/91.33	70.04/82.38	70.54/82.73

TABLE VI
COMPARISON OF COMPUTATIONAL COMPLEXITY AND PARAMETERS

Methods	Pub.	FLOPs (G)	Params (M)
S4GAN [30]	TPAMI19	73.23	46.85
SemiCDNet [42]	TGRS20	73.24	46.85
SemiCD [43]	Arxiv22	73.24	46.85
RCL [50]	TGRS22	73.23	46.85
UniMatch [33]	CVPR23	72.25	42.88
FPA [51]	TGRS23	73.23	46.85
ECPS [47]	TGRS24	**10.85**	**3.71**
CutMix-CD [44]	TGRS25	73.23	46.85
AdaSemiCD [48]	TGRS25	73.23	46.85
UniMatch-V2 [29]	TPAMI25	97.50	137.65
DistillSemiCD	—	79.76	53.89

under different values of m, indicating that it is not highly sensitive to this parameter. Overall, $m = 0.3$ achieves the most balanced results, obtaining the best IoU on WHU-CD under the 10% labeled setting and on SYSU-CD under both 5% and 10% labeled settings. When m is too small, the interclass separation constraint may be insufficient, while an overly large m may disturb intraclass compactness and slightly degrade performance. Therefore, we set $m = 0.3$ in all experiments to balance feature discriminability and training stability.

F. Complexity Analysis

Table VI reports the computational complexity and parameter size of the compared methods. DistillSemiCD requires 79.76G FLOPs and 53.89 M parameters during inference. Since the DINOv2 teacher is used only in the training stage and removed during inference, the deployed model remains a compact student network rather than a foundation-model-based detector. Compared with UniMatch-V2, which requires 97.50G FLOPs and 137.65 M parameters, DistillSemiCD reduces FLOPs by approximately 18.2% and parameters by approximately 60.9%. Although UniMatch-V2 achieves higher accuracy in several WHU-CD and LEVIR-CD settings, its larger model size leads to a heavier deployment burden. On the contrary, DistillSemiCD maintains competitive performance with substantially fewer parameters, demonstrating a more balanced tradeoff between accuracy and efficiency.

TABLE VII
CROSS-DOMAIN SEMISUPERVISED PERFORMANCE. RED AND BLUE INDICATE THE BEST AND SECOND-BEST RESULTS. † DENOTES RESULTS CITED FROM [43]; * DENOTES RESULTS REPRODUCED UNDER IDENTICAL PROTOCOLS

Method	{LEVIR (sup.%), WHU unsup.} → LEVIR-CD			
	5% IoU/F1	10% IoU/F1	20% IoU/F1	40% IoU/F1
AdvNet†	65.20/78.93	71.60/83.45	75.00/85.71	76.50/86.69
S4GAN†	66.70/80.02	68.40/81.24	73.40/84.66	75.30/85.91
SemiCDNet†	67.60/80.67	71.90/83.65	74.90/85.65	76.40/86.62
SemiCD†	71.40/83.31	74.60/85.45	76.40/86.62	77.90/87.58
FPA*	58.41/73.75	66.26/79.71	70.61/82.77	68.02/80.97
UniMatch*	67.37/80.50	70.82/82.92	74.87/85.63	76.52/86.70
SemiBCD*	**77.02/87.02**	78.69/88.07	79.55/88.61	80.19/89.01
UniMatch-V2*	**76.65/86.78**	**79.39/88.51**	**80.65/89.29**	**81.31/89.69**
Ours	74.98/85.70	**79.88/88.81**	**80.34/89.10**	**80.79/89.37**

G. Cross-Domain Generalization

To further validate the robustness of DistillSemiCD under cross-domain scenarios, we conduct experiments following the protocol established in [43]. In addition to the reported classical baselines, we reproduce SemiBCD and UniMatch-V2 under the same protocol for a more comprehensive comparison. As illustrated in Table VII, DistillSemiCD achieves the best IoU at the 10% labeled ratio with 79.88%, and ranks second in IoU at the 20% and 40% labeled ratios with 80.34% and 80.79%, respectively, only marginally below UniMatch-V2. At the 5% labeled ratio, SemiBCD and UniMatch-V2 achieve higher IoU/$F1$ scores than DistillSemiCD. Nevertheless, DistillSemiCD consistently outperforms the classical baseline SemiCD across all labeled ratios, with particularly clear improvements in IoU. These results suggest that DistillSemiCD can effectively leverage heterogeneous cross-domain unlabeled data. This improvement can be attributed to the DINO-APC module, which exploits domain-agnostic structural priors from DINOv2 to alleviate pseudolabel noise caused by domain shift, and to the batch-level prototype alignment in DLCL, which helps maintain more stable feature distributions when unlabeled data originates from a different domain.

V. CONCLUSION

This article proposes DistillSemiCD, a semisupervised RSCD framework that efficiently deploys foundation-model knowledge while mitigating pseudolabel noise and boundary ambiguity. The TID-KD strategy transfers change-aware representations from a frozen DINOv2 teacher to a lightweight student, complemented by DINO-APC, DLCL, and TIBR for enhanced pseudolabel reliability, feature discriminability,

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and boundary refinement. Experiments on four benchmarks validate the framework's effectiveness.

Nevertheless, limitations remain: the domain gap between DINOv2 pretraining data and remote sensing imagery may constrain fine-grained change sensitivity, and TIBR introduces additional computational overhead. Future work will explore remote sensing-specific foundation models, robust cross-domain adaptation, and multimodal change detection combining optical and SAR data.

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